Thermodynamic description of the Cu–Mg system: a synthetic benchmark document

ORCHID synthetic benchmark (generated; no human authors)

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Abstract

This synthetic document describes the Cu–Mg binary system as calculated with pycalphad from the SGTE dataset CuMg-16Hal.tdb; every figure, table and stated value is generated, and the exact values are kept in a separate truth key. The highest invariant reaction, L → (Cu) + Cu2Mg, lies at 725.2 °C.

1. Introduction

The Cu–Mg system belongs to the 20 binaries of the ORCHID benchmark, chosen to cover diagram types such as congruent compound, intermetallic rich. The thermodynamic description is the SGTE collection entry (Hallstedt, 2025), which reproduces the assessment of B. Hallstedt, unpublished, 2016. The calculations use pycalphad 0.11.2 at 1 atm.

Pure Cu melts at 1084.6 °C. The melting point of Mg reproduced by the dataset is 649.9 °C.

2. Phase diagram

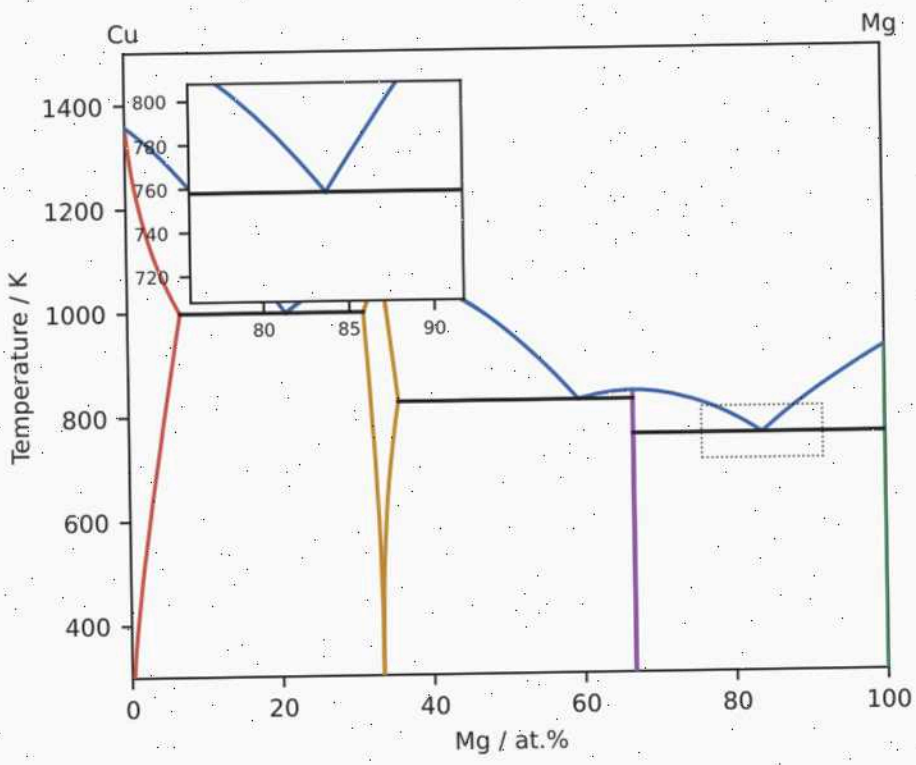


Fig. 1. Calculated Cu–Mg phase diagram (inset: enlarged view near the lowest invariant).

At 725.2 °C the eutectic L → (Cu) + Cu2Mg takes place; the L phase then contains 20.9 at.% Mg. The dataset places a eutectic (L → Cu2Mg + CuMg2) at 552.0 °C; the composition of L is 59.5 at.% Mg. Eutectic equilibrium L → CuMg2 + (Mg) is found at 484.9 °C and 83.7 at.% (L).

Table 1. Invariant reactions in the Cu–Mg system (calculated).

Reaction	Type	T (K)	at.% Mg, 1	at.% Mg, 2	at.% Mg, 3
L → (Cu) + Cu2Mg	eutectic	998.4	20.9	7.0	31.2
L → Cu2Mg + CuMg2	eutectic	825.2	59.5	35.7	66.7
L → CuMg2 + (Mg)	eutectic	758.1	83.7	66.7	100.0

Phases 1–3 in the order written in the reaction. Calculated with pycalphad 0.11.2 from the SGTE dataset; synthetic document.

3. Thermodynamic properties

At 1450 K the integral mixing enthalpy of the liquid has its minimum, −9.35 kJ/mol, at 44.0 at.% Mg. The calculated enthalpy of formation of CuMg2 at 298.15 K is −8.65 kJ/mol per mole of atoms.

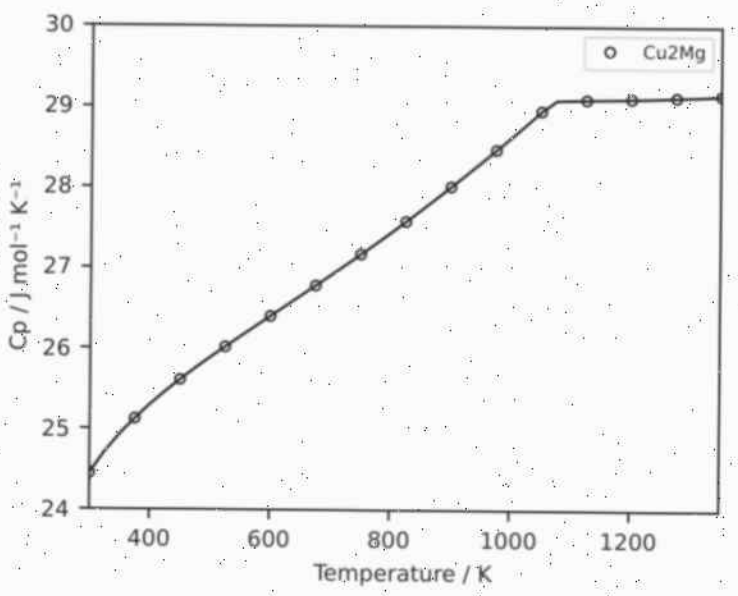


Fig. 2. Calculated heat capacity of Cu2Mg.

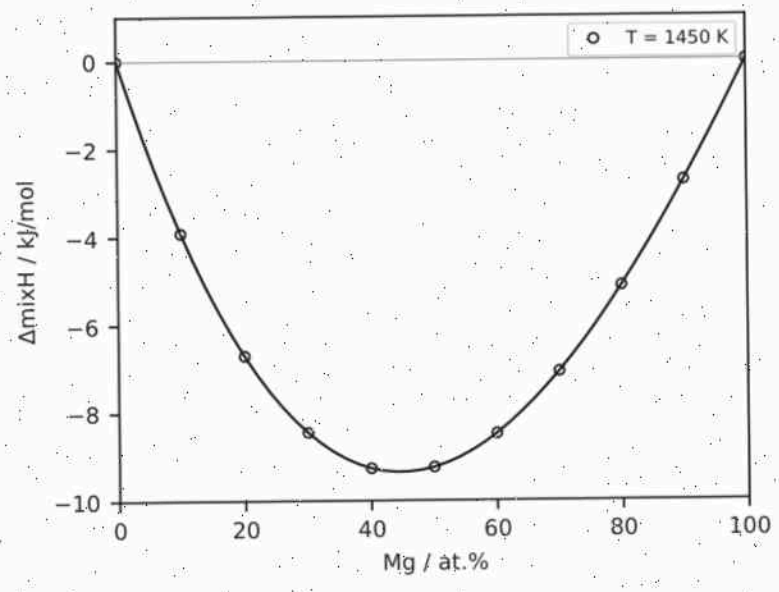


Fig. 3. Calculated mixing enthalpy of the liquid at 1450 K.

Table 2. Calculated formation properties of the intermediate phases (Cu–Mg).

Phase	at.% Mg	ΔfH(298 K) / kJ mol⁻¹	S(298 K) / J mol⁻¹ K⁻¹
Cu2Mg	33.5	-10.20 ± 0.5	34.95
CuMg2	66.7	-8.65 ± 0.5	33.50

Reference: stable elements (SER) at 298.15 K. ± values are synthetic illustration uncertainties, not data.

4. Conclusions

This document is synthetic and exists only to test extraction tools; its truth key lists every boundary, field, invariant, plotted point, table cell and stated value.